

PD_SMARTKEY_EMN-EVAN_V2

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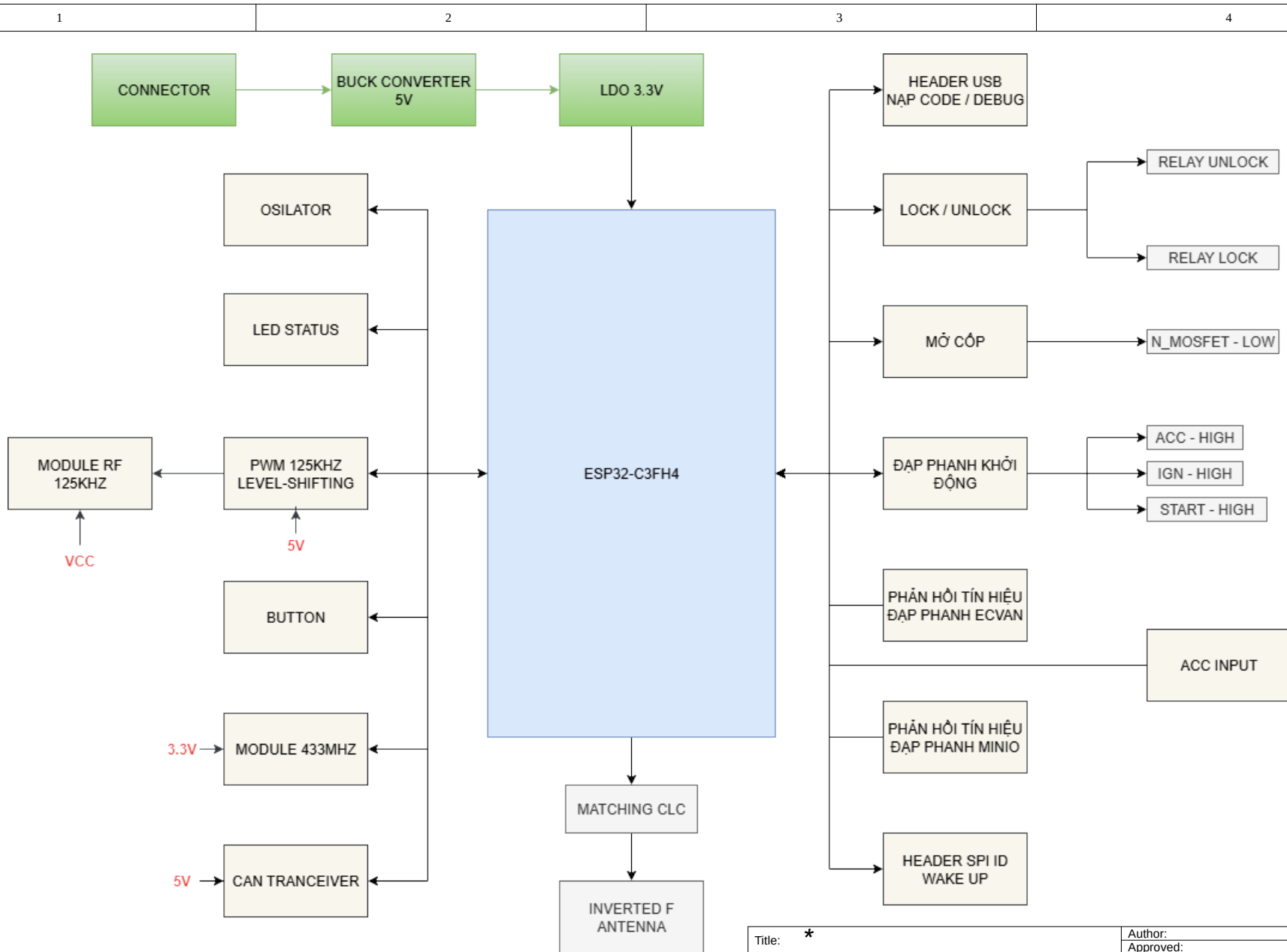
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DESIGN CONSIDERATIONS

DESIGN NOTE:
Functions of this block

DESIGN NOTE:
Calculation equations

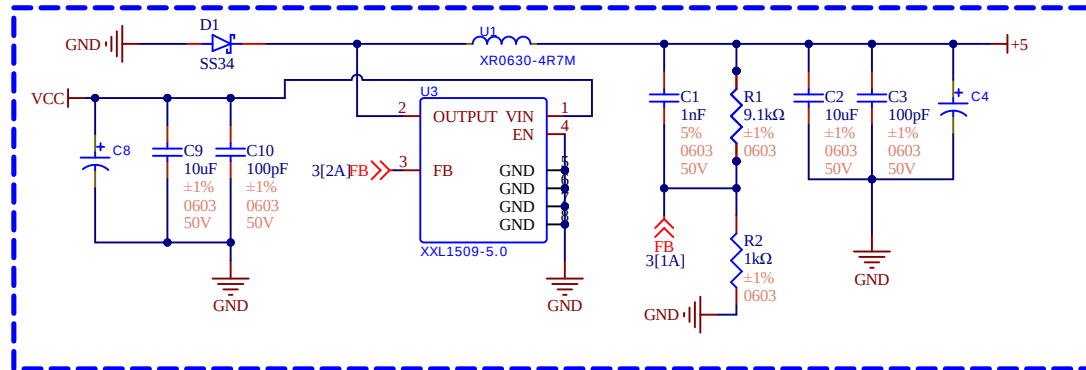
DESIGN NOTE:
Calculations when choosing this component



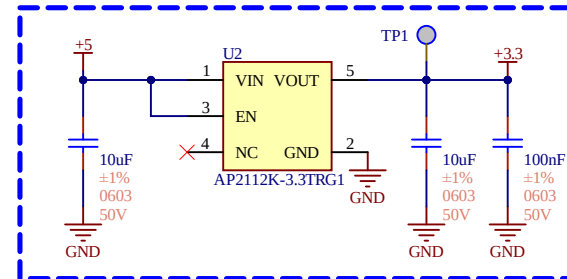
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Size: A4		Approved: *	
Project: PD_SMARTKEY_EMN-EVAN_V2		CONFIDENTIAL	
Date: 9/3/2026		Modified Date: 8/12/2026	
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BUCK CONVERTER 5,9V



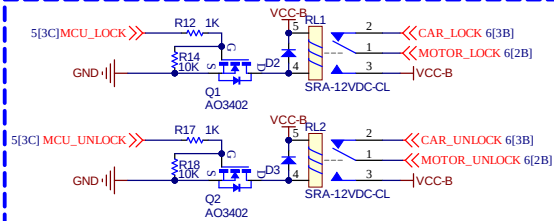
LDO 3.3V



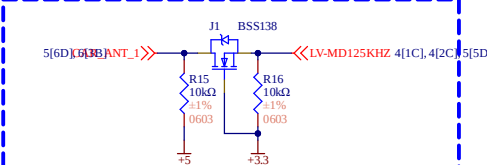
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Date: 9/3/2026		Revision: *	
5:52:16 PM		SW Version: 24.1.2.44	
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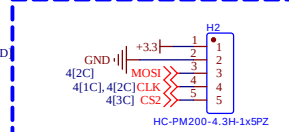
LOCK / UNLOCK



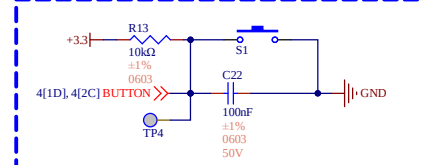
LEVEL SHIFTING



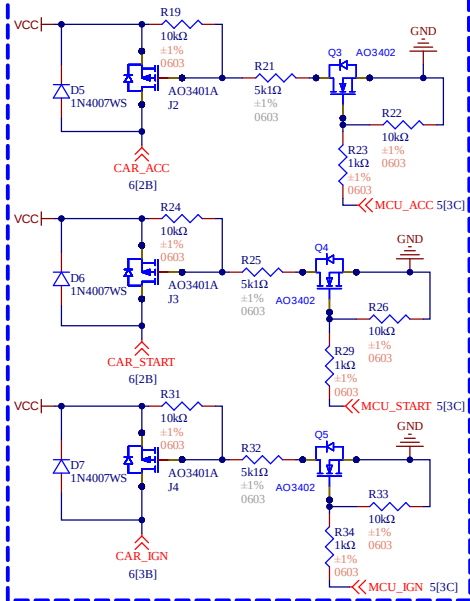
SPI ID WAKE UP



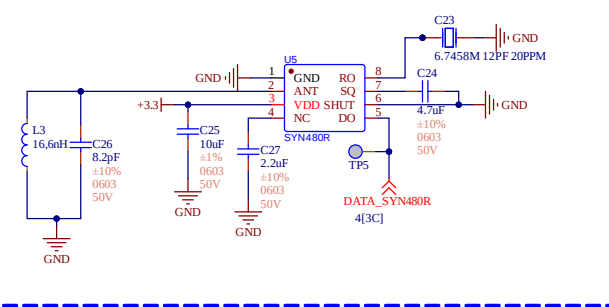
BUTTON



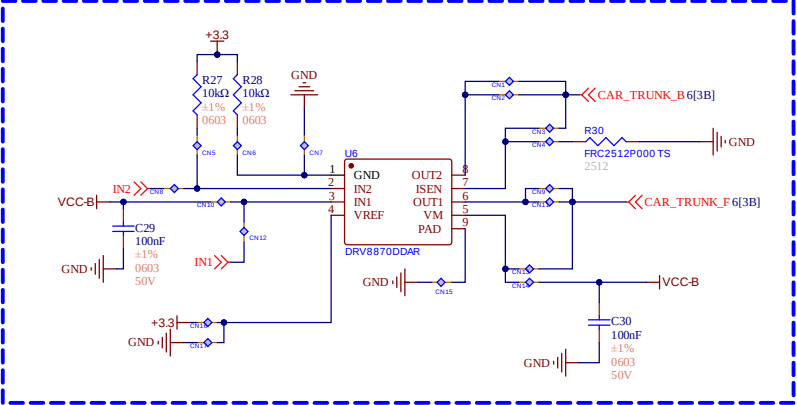
KHOẢNG ĐỘNG XE



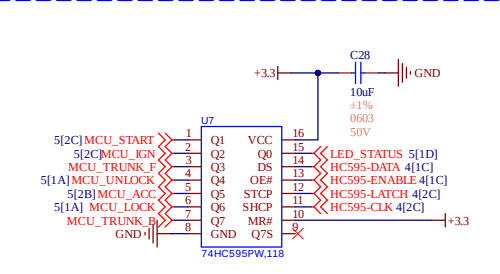
433Mhz



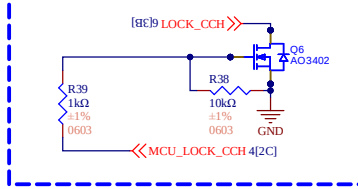
DRIVER MOTOR



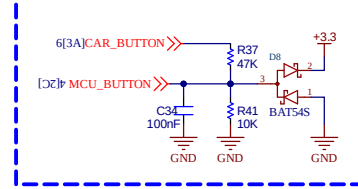
IC RA CHÂN



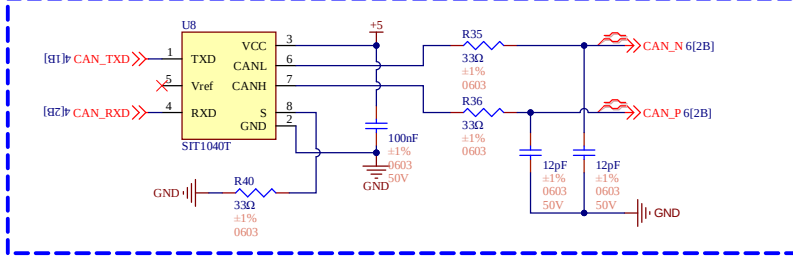
CENTER LOCK



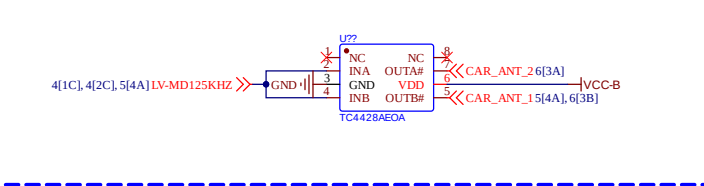
BUTTON



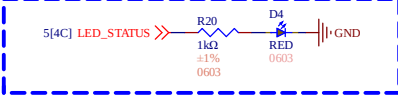
CAN TRANSCEIVER



ANT-125kKHZ

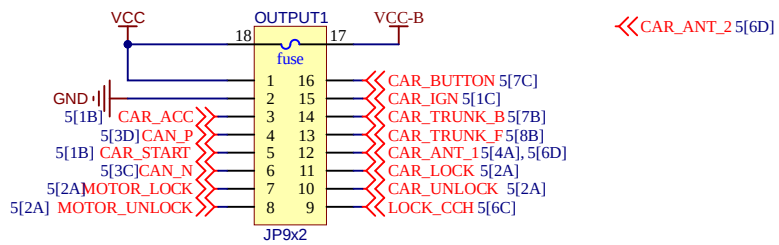


LED STATUS



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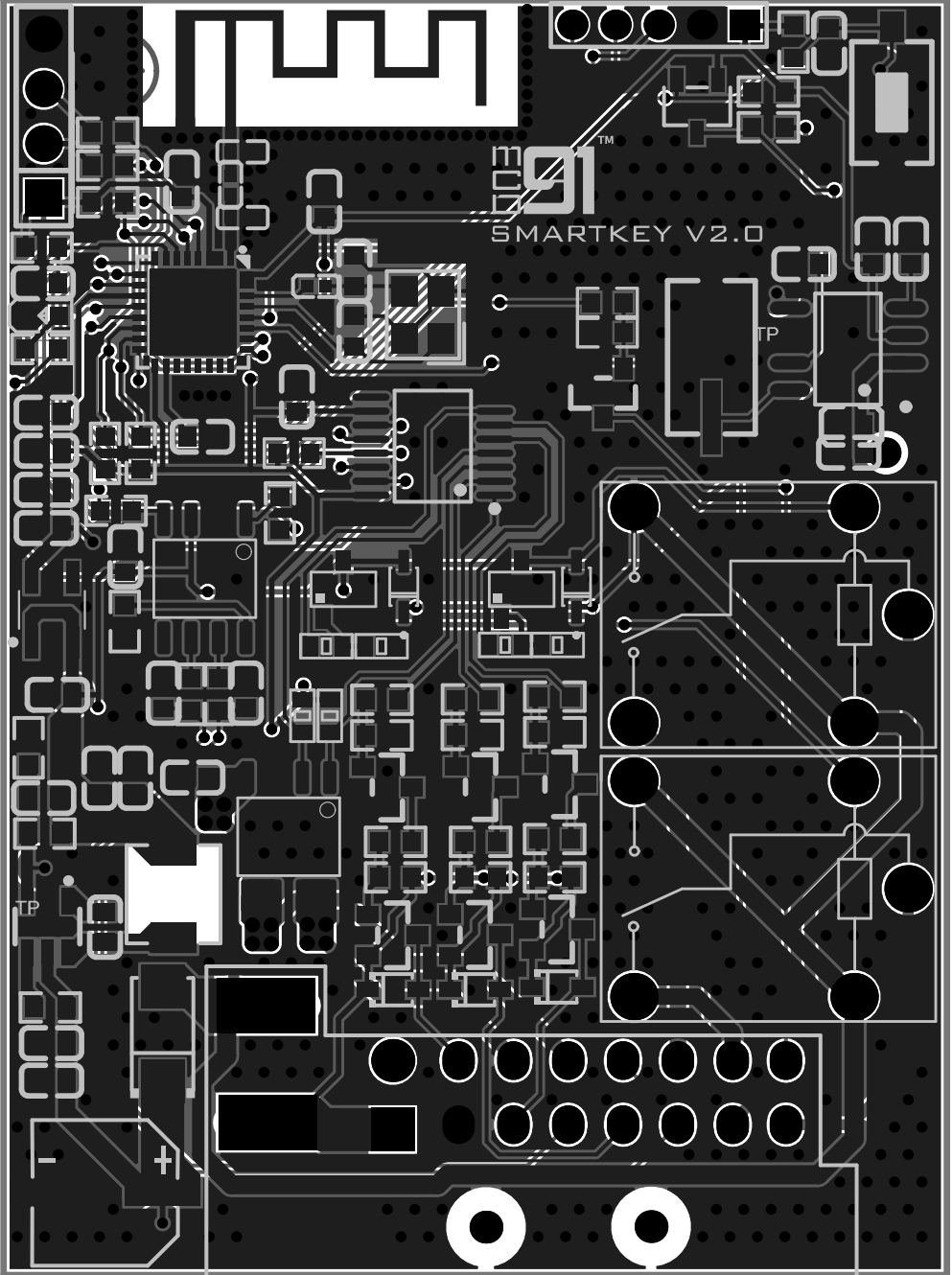


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			Approved: *
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CONFIDENTIAL

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Description	Designator	Quantity	Package	Manufacturer	Mfr. Part #	MPN
100pF 50V 100pF NPO 41% 0402 Multilayer Ceramic Capacitors MLCC - SMD/SMT 100V	C1	1	1003			
100pF 50V 100pF NPO 41% 0402 Multilayer Ceramic Capacitors MLCC - SMD/SMT 100V	C2, C8, C9, C8, C11, C13, C25, C28	8	0603	YAGEO	CC0402F1NP00B N100	
100pF 50V 100pF NPO 41% 0402 Multilayer Ceramic Capacitors MLCC - SMD/SMT 100V	C11, C16	2	0603	YAGEO	CC0402F1NP00B N100	
CAP ALUM 220UF 20% 50V TH	D4, C8	2	Through Hole/D204.16mm	ADPH	BRUHM225160T	
50V 100pF NPO 41% 0402 Multilayer Ceramic Capacitors MLCC - SMD/SMT 100V, 100V	C7, C14, C21, C22, C29, C30, C31	7	0603	YAGEO	CC0402F1NP00B N100	
50V 100pF NPO 41% 0402 Multilayer Ceramic Capacitors MLCC - SMD/SMT 100V	C11, C16	2	0603	YAGEO	CC0402F1NP00B N100	
Crystal Oscillator 10pF 10000000-4P Crystals 10MHz	C15	1	14MD3225-4P	HYC Crystal Oscillators	RL2EL80CR-1111LC-40M	
50V 100pF NPO 41% 0402 Multilayer Ceramic Capacitors MLCC - SMD/SMT 100V	C17, C18	2	0402	YAGEO	CC0402F1NP00B N100	
50V 100pF NPO 41% 0402 Multilayer Ceramic Capacitors MLCC - SMD/SMT 100V	C18, C20, C32, C33	4	0603	YAGEO	CC0402F1NP00B N100	
CRYSTAL 6.7458MHz 10pF SMD	C23	1	HC-49S SMD-2P	COSET	6.7458M 12PF 100MA	
4.7uF	C24	1	0603	muRata	GCJ188C701050KE 100V	
0.2uF	C26	1	0603	muRata	GCJ188C701050KE 100V	
2.2uF	C27	1	0603	muRata	GCJ188C701050KE 100V	
Cap SMD	C28	1				
	CN1, CN2, CN3, CN4, CN5, CN6, CN7, CN8, CN9, CN10, CN11, CN12, CN13, CN14, CN15, CN16, CN17	17				
40V Independent 3A 900mV@3A 1MA@0.1V4AC LowRds MOSFET 3045	D1	1	1MA@0.1V4AC	MCO/Microdiode Semiconductor	1034	
Light Bulb	D2, D3	2				
Clear 40°C~105°C 60mW 0402 LED Indicator - Discrete 100V	D4	1	0603	WINGLIGHT	WL1005UC	
DIODE STANDARD 3AV LA SMD-323	D5, D6, D7	3	300-323	MCO/Microdiode Semiconductor	1N4007WS	
Diode 50V 0.2A Surface Mount SOT-23	D8	1				
DAV 1.254mm Pin Header 5x2.54mm Through Hole 5mm - 40°C~105°C 5mm Gold 4P Black 144P Through Hole P-2.54mm Headers, Male Pins 100V	H1	1		LITTECON	P12 54 104P-A60	
CONN SOCKET SP08 8pin 180G E ROW TH MICROFIT N CH 30V 0.22A SOT-23	J2	1		Hong Cheng	HC-PAD008-4.3H 149P2	
20V 4A 900mV@250A 1-P Channel SOT-23 MOSFET 100V	J3	1	1017-23	phoenix	88513B	
20V 4A 900mV@250A 1-P Channel SOT-23 MOSFET 100V	J2, J3, J4	3	3017-23	ACS	A02401A	
FIRED IND 14H 1A 12mV@100V	L1	1		muRata	LQG15H43M0800	
FIRED IND 24H 1A 20mV@100V	L2	1	0402	muRata	LQW15AN24M000 0	
70mH 50mV 40% 170mH Wire wound Inductor 0603 (Fixed) Inductors	L3	1	0603	Collark	HWCL100116M3	
WR-PH2 2.54 mm Quad Invert Header	OUTPUT1	1		W" 100 Development 25000 Copper		
N Channel 20V 2.5A 1% 80mW Tc Surface Mount TO-18 250V SOT-23	Q1, Q2	2	new			
MICROFIT N CH 30V SOT-23	Q3, Q4, Q5, Q6	4		ACS	A03403	
0.1uF	R1	1	0603	YAGEO	RC0402FR-0710R	
02 50mW Thick Film Resistor 50V ±1% ±200ppm/°C 100 0402 Chip Resistor - Surface Mount 100V	R2, R20, R21, R23, R24, R25	6	0603	YAGEO	RC0402FR-0710R	
02 50mW Thick Film Resistor 50V ±1% ±200ppm/°C 100 0402 Chip Resistor - Surface Mount 100V	R3, R6, R9, R10, R11, R13, R15, R16, R19, R22, R24, R26, R27, R28, R31, R33, R38	17	0603	YAGEO	RC0402FR-0710R	
100mW Thick Film Resistor 75V ±5% 50 0603 Chip Resistor - Surface Mount 100V	R4, R5	2	0603	YAGEO	RC0603JR-070R	
100mW Thick Film Resistor 75V ±1% ±200ppm/°C 10 0603 Chip Resistor - Surface Mount 100V	R6, R7, R35, R36, R40	5	0603	YAGEO	RC0603FR-072R	
Resistor	R12, R17	2		Generic Components		
Resistor	R14, R18, R41	3		Generic Components		
100mW Thick Film Resistor 75V ±100ppm/°C ±1% 5 340 0603 Chip Resistor - Surface Mount 100V	R21, R25, R32	3	0603	ROJAN	PRC0603FS101T5	
RES-01 10% 1W 2512	R30	1		COJAN	RC012512W001T5	
Resistor	R37	1		Generic Components		
Relay SPDT, 10A - 12VDC 1-in.	RF1	1				
Switch	RL1, RL2	2				
	RF1, RF2, RF3, RF4	4				
FIRED IND 6.7458MHz 10pF SMD	U1	1		AR	XR0630-407M	
1-Pin Voltage Regulator C Positive Fixed 1 Output 600mA SOT-23	U2	1	107-173 Diodes etc.			
150kHz 3A PWM Buck DC/DC Converter	U3	1		HMW	KLL1500-5.0	
Other 150kHz 3A 30-EP(S) RF Transceiver Modules and Modems RoHS	U4	1	QFN-32 4P(S)4	ESPRESBF	ESP32-C3	
433/315 MHz wireless receiver chips	U5	1		LSM8EM	DTN480R	
Integrated IC motor driver PWM control	U6	1		TI	DRV8870CDAR	
8 bit serial in, serial or parallel out shift register with output latches, 3-state	U7	1		Winperla	74HC595PW118	
171 Transceiver HBR C4040 8-20	U8	1	101C-8 NXP USA			
1.5A Dual High Speed Power MOSFET Drivers	U77	1		MICROCHIP	TC4428AECOA	